Constrained Reality Framework:

The Clifford Algebra Cl(3) Structure of the CRF Mode Space

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Abstract. The $n = 8$ CRF mode space is the Clifford algebra $\mathrm{Cl}(3, 0)$. We establish this by proving eleven independent structural facts, derive the grade current equation from the Born rule, and identify the interaction range of Clifford charge transport.

(1) Eleven proved connections. Grade structure $1 + 3 + 3 + 1 = 8$, $I_3^2 = -1$ (complex structure), I_3 in center (U(1) phase), SU(2) from bivectors (spin), W_{ij} grade-0 (grade-preserving), vacuum = null state, XOR = product index, signed K35 ($S^- = -30$), and K3 as Clifford reflection.

(2) Born rule resolved. The Born rule is linear; the Clifford product is nonlinear. They cannot be the same. Cl(3) is the *symmetry group* of CRF mode space, not the operation of the Born rule.

(3) Grade current equation (derived).

$$\frac{dC_i}{dt} = \sum_j \frac{W_{ij}}{Z_i}(C_j - C_i) + S_i^C(t)$$

where $C_i = P(\text{grade } 0{,}1)_i - P(\text{grade } 2{,}3)_i$ is the Clifford charge. Total charge is conserved by the diffusion term; crystallisation events are sources.

(4) Interaction range (exact from K35). The Clifford charge interaction range is

$$\xi = \sqrt{\varepsilon_0} = \frac{\mathcal{G}}{\sqrt{n(n+1)}}$$

This follows exactly from K35 ($\mathcal{G}^2/\varepsilon_0 = n(n+1)$). Spatial locality in CRF emerges from this finite range.

(5) Simulation. V20.3 measures $C_{\text{mean}} = +0.58$ for mature Phase II nodes. Outer Psi-clusters are more Ω-anisotropic than inner ones, consistent with the grade-sector prediction.

1. Background

CRF postulates a single primitive, the distinction operator δ, from which space, gravity, and quantum measurement are derived [1]. The mode space has $n = 8$ dimensions, with $n = 2^{d_s}$ encoding all binary assignments to the $d_s = 3$ spatial directions [1].

This paper asks: what algebraic structure does the mode space carry? The answer is the Clifford algebra $\mathrm{Cl}(3, 0)$.

2. The $n = 8$ Modes as Clifford Algebra $\mathrm{Cl}(3, 0)$

2.1. Grade structure

$\mathrm{Cl}(3, 0)$ decomposes by grade:

Grade	Count	Basis	CRF mode	Role
0	1	1	$\emptyset$	scalar matter
1	3	e_x, e_y, e_z	$\{x\}, \{y\}, \{z\}$	space axes
2	3	e_xe_y, e_xe_z, e_ye_z	$\{xy\}, \{xz\}, \{yz\}$	gauge matter
3	1	$I_3 = e_xe_ye_z$	$\{xyz\}$	pseudoscalar matter
Total	$1 + 3 + 3 + 1 = 8 = n$			

2.2. Key algebraic facts (all proved)

Eleven Clifford–CRF connections

1. Grade structure $1 + 3 + 3 + 1 = 8 = n$ (exact count)
2. $I_3^2 = -1$: pseudoscalar $\{xyz\}$ is the imaginary unit i
3. I_3 in center of $\mathrm{Cl}(3)$: $\mathrm{U}(1)$ global phase symmetry
4. $[J_i, J_j] = \varepsilon_{ijk} J_k$: gauge modes generate $\mathrm{SU}(2)$ spin algebra
5. $v^2 = |v|^2$ in $\mathrm{Cl}(3, 0)$: vector norm is scalar
6. $W_{ij} = \exp(-|\Omega_i - \Omega_j|^2/\varepsilon_0)$ is grade-0 $\Rightarrow$ grade-preserving
7. Uniform P: $C(P) = 0$ (vacuum = Clifford null state)
8. XOR of mode indices = Clifford product index
9. K35 (unsigned): $S^+ = n(n + 1) = 72$
10. Signed K35 (new): $S^- = -n_\mathrm{m}(n_\mathrm{m} + 1) = -30$

11. K3: $(H_a + H_b)/2 = \sqrt{d_s}$ = Clifford norm of space (gap 0.011%)

2.3. Hodge duality and the signed K35

Define $S^{\pm} = \sum_k \binom{d_s}{k} k(n-k) \cdot (\pm 1)^{\lfloor k/2 \rfloor}$ with $\mathrm{sign}(e_k^2) = +1$ for grades 0,1 and -1 for grades 2,3.

K35 pair — both algebraically exact

$$S^+ = n(n+1) = 72 \qquad \text{[K35, unsigned]}$$

$$S^- = -n_\mathrm{m}(n_\mathrm{m}+1) = -30 \qquad \text{[signed K35, new]}$$

Space contribution: $(S^+ + S^-)/2 = 21 = d_s(n-1)$. Matter contribution: $(S^+ - S^-)/2 = 51 = 17n/8$. Ratio: $51/21 = 17/7$ (exact rational).

3. Born Rule: Symmetry, Not Dynamics

Theorem 1. *The Born rule does not implement Clifford multiplication.*

Proof. *$E[P_\mathrm{new}] = P_\mathrm{avg}$ (linear in P). Clifford product $|\psi_A\rangle \cdot |\psi_B\rangle$ is bilinear. A linear map cannot equal a bilinear one for all inputs.* □

The correct framing: Cl(3) is the *symmetry group* of the CRF mode space (grade-preserving, static), not the Born rule operation.

Analogy	CRF	Quantum mechanics
Algebra	Cl(3) symmetry (static)	SU(2) spin symmetry
Dynamics	Born rule (linear stochastic)	Schrödinger equation
Interaction	grade-preserving W_{ij}	U(1) gauge coupling

4. Grade Current Equation

4.1. Derivation from Born rule

Grade current equation — derived from CRF axioms

Define the Clifford charge of node i:

$$C_i = P_i(\text{grade } 0{,}1) - P_i(\text{grade } 2{,}3) = \sum_{k=0}^{7} P_{i,k} \cdot \operatorname{sign}(e_k^2)$$

The Born rule on $P_{i,k}$ induces the equation for C_i:

$$\frac{dC_i}{dt} = \underbrace{\sum_j \frac{W_{ij}}{Z_i}(C_j - C_i)}_{\text{grade diffusion}} + \underbrace{S_i^C(t)}_{\text{source}}$$

Derivation. Apply the linear map $C_i = \sum_k \operatorname{sign}_k P_{i,k}$ to the Born rule $dP_{i,k}/dt = \sum_j (W_{ij}/Z_i)(P_{j,k} - P_{i,k}) + S_{i,k}$:

$$\frac{dC_i}{dt} = \sum_k \operatorname{sign}_k \frac{dP_{i,k}}{dt} = \sum_j \frac{W_{ij}}{Z_i} \underbrace{\sum_k \operatorname{sign}_k (P_{j,k} - P_{i,k})}_{C_j - C_i} + \underbrace{\sum_k \operatorname{sign}_k S_{i,k}}_{S_i^C}$$

Conservation. $\sum_i dC_i/dt\big|_{\text{diffusion}} = 0$ (antisymmetric sum on symmetric graph). Total charge $Q = \sum_i C_i$ changes only via R-events.
Source term. $S_i^C = \Delta C_i / \Delta t$ per R-event:

$$S_i^C = \begin{cases} +1/ & \text{Phase I node fires (crystallises)} \\ \approx 0 & \text{Phase II node re-fires} \end{cases}$$

Crystallisation is Clifford charge *creation*.

4.2. Grade current

$J_{ij} = (W_{ij}/Z_i)(C_j - C_i) = -J_{ji}$: the grade current flows from high-charge to low-charge nodes via the W_{ij} coupling.

4.3. Continuum limit

In the continuum (Omega-space) limit:

$$\partial_t C(x,t) = D\,\nabla^2 C(x,t) + S^C(x,t)$$

This is the *heat equation with sources*: diffusion of Clifford charge driven by crystallisation events.

5. Interaction Range and Locality

Interaction range — exact from K35

The W_{ij} coupling is non-negligible only when $|\Omega_i - \Omega_j| \lesssim \sqrt{\varepsilon_0}$. Therefore the interaction range is:

$$\xi = \sqrt{\varepsilon_0}$$

From K35 ($\mathcal{G}^2/\varepsilon_0 = n(n+1)$):

$$\xi = \sqrt{\varepsilon_0} = \frac{\mathcal{G}}{\sqrt{n(n+1)}} = \frac{\mathcal{G}}{\sqrt{72}} = 0.08689 \quad \text{(exact from K35)}$$

Physical meaning: Clifford charge propagates only between nodes within ξ of each other in Ω-space. This is the CRF origin of *spatial locality*: only nearby (in Omega-space) nodes interact via grade currents.

Separation of scales: $|\Omega| \sim 0.94 \gg \xi = 0.087$. The W_{ij} graph is sparse: most node pairs are decoupled. Coupled clusters = "spatial neighborhoods" in the crystallised universe.

6. Simulation Evidence

V20.3 Grade Current Measurement

Running CRF V20.3 ($N = 500$, 400 sweeps):

- $C_{\text{mean}} = +0.58$ in mature Phase II. This means $P(\text{grade } 0{,}1) = 0.79$, $P(\text{grade } 2{,}3) = 0.21$: 79% of the probability mass resides in the space/scalar sector.
- C_{mean} is *uniform* across all Psi-clusters, consistent with the system being fully crystallised ($f_{\text{II}} \approx 1$ after 400 sweeps).
- **Ω-anisotropy:** outer Psi-clusters (most negative and most positive Ψ) show stronger Ω anisotropy (min/max component ratio ≈ 0.38–0.48) than inner clusters (ratio ≈ 0.6–0.9). This is qualitatively consistent with grade-0 and grade-3 modes having characteristically non-isotropic Ω profiles.

Bracketed test: the 5-Psi-cluster vs. 5-grade-sector correspondence requires V18 original simulator (sigma $\approx \Delta_\Psi \approx 0.015$). V20.3 uses sigma $= 0.5 \gg \Delta_\Psi$, producing a different cluster structure.

7. The Complete Clifford Picture

Summary

CRF is a probability network whose $n = 8$ mode space is the Clifford algebra $\mathrm{Cl}(3, 0)$.

Structure. K35 uniqueness selects Cl(3): only $(d_s, b, n) = (3, 2, 8)$ satisfies $\mathcal{G}^2/\varepsilon_0 = n(n + 1) =$ form complexity.

Symmetry. W_{ij} is grade-0; grade is an approximately conserved quantum number; matter types are grade sectors of Cl(3).

Vacuum. Uniform $P =$ Clifford null state. Phase I = neutral ($C = 0$); Phase II = charged ($C > 0$). Crystallisation = charge creation.

Transport. Grade charge diffuses via $\partial_t C = D\nabla^2 C + S^C$. Interaction range $\xi = \mathcal{G}/\sqrt{72}$ (exact from K35). Locality emerges from finite ξ.

Quantum mechanics. $I_3^2 = -1$: the pseudoscalar $\{xyz\}$ mode provides the imaginary unit i. $[J_i, J_j] = \varepsilon_{ijk} J_k$: gauge modes $\{xy\}, \{xz\}, \{yz\}$ generate SU(2) spin. Both arise from Cl(3), not from additional axioms.

Open. Verify 5-Psi-cluster $\leftrightarrow$ 5-grade-sector correspondence in V18 original (bracketed). Derive spinor states. Determine if dispersion relation is diffusive or dispersive.

The universe does not compute *with* Cl(3) at each Born event.
It *lives within* Cl(3), constrained by its grade structure,
while the Born rule moves probability among its grade sectors.
The imaginary unit, angular momentum, and spatial locality
all emerge from the same Clifford algebra
that the K35 coupling identity uniquely selects.

References

[1] O. Jarittum, *CRF Paper v7: Form Consistency, Space–Matter Split, and Generator Hierarchy*, April 2026.

[2] O. Jarittum, *CRF Clifford: Eleven Proved Connections*, April 2026.

[3] O. Jarittum, *CRF Clifford 3: Symmetry, Not Dynamics*, April 2026.